Foundations of Programming

Assignment No. : 12

PROBLEM STATEMENT:

Write a program in C to accept two numbers from the user and compute:

a) The smallest common divisor of the two numbers (other than 1, if any)

b) The Greatest Common Divisor (GCD) of the two numbers

OBJECTIVE:

• To understand the use of mathematical operations in C.

• To implement loops and conditional statements

. • To perform number-based computations.

• To enhance logical thinking using menu-driven programs.

THEORY:

Divisors and the Greatest Common Divisor (GCD) are fundamental concepts in number theory. The smallest common divisor of two numbers is the smallest positive integer greater than 1 that divides both numbers. The GCD of two numbers is the largest positive integer that divides both numbers without leaving a remainder. In C programming, these operations can be efficiently implemented using loops and the Euclidean algorithm, which repeatedly applies the modulo operation until the remainder becomes zero.

PLATFORM:

Linux

ALGORITHM:

C PROGRAM:

```
#include<stdio.h>


int gcd(int a,int b)

{

    while(b!=0)

    {

        int t=b;

        b=a%b;

        a=t;

    }

    return a;

}


int smallestCommonDivisor(int a,int b)

{

    int min = (a<b)?a:b;

    for(int i=2;i<=min;i++)

        if(a%i==0 && b%i==0)

            return i;

    return -1;   // no divisor other than 1

}
```

```c
int main()
{
    int a,b;
    printf("Enter two numbers: ");
    scanf("%d%d",&a,&b);

    int scd = smallestCommonDivisor(a,b);
    int g = gcd(a,b);

    if(scd==-1)
        printf("No common divisor other than 1\n");
    else
        printf("Smallest Common Divisor = %d\n",scd);

    printf("GCD = %d\n",g);
}
```

INPUT:

```
Enter two numbers: 12 18
```

OUTPUT:

```
Smallest Common Divisor = 2
GCD = 6
```

CONCLUSION:

Thus, the program to find the smallest common divisor and the Greatest Common Divisor of two numbers was successfully implemented using C programming. This program demonstrates the use of loops, conditional statements, and the Euclidean algorithm.

FAQs:

1. What is meant by the Greatest Common Divisor (GCD)?
2. Why does the Euclidean algorithm efficiently compute the GCD?
3. How can this program be modified to handle negative numbers?
4. What changes are required to compute the Least Common Multiple (LCM) using the GCD?
5. How can this program be extended to find the LCM of two numbers?